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DBMS

Assignment-1

Sec-A Introduction to DBMS

Q1 What is a Database Management System (DBMS)? Why is it required?

Ans A DBMS is a software that enables users to define, create, maintain & control access to a database.

Why required: Controls data redundancy, maintains data consistency, provides security, ensures data integrity.

Q2 Why is DBMS better than the traditional file system?

Ans DBMS was centralized control, whereas file systems store data separately in files. It reduces redundancy, controls consistency, access control, Back-up & recovery.

Q3 What is data abstraction? Explain its level.

→ DBMS abstraction hides storage details from users
Three levels

- ↳ Physical level: How data is stored (index, storage blocks)
- ↳ Logical level: What data is stored.
- ↳ View level: What users sees.

Q4 What is Three-schema architecture?

- External level
- Conceptual level
- Internal level.

Q5 What is data independence?

- ability to modify schema at one level without affecting other level
- ↳ Physical data independence.
- ↳ logical

Q6 What is ER mode?

- It is a diagram that represents entities, attributes, relationships. It is used for database design planning.

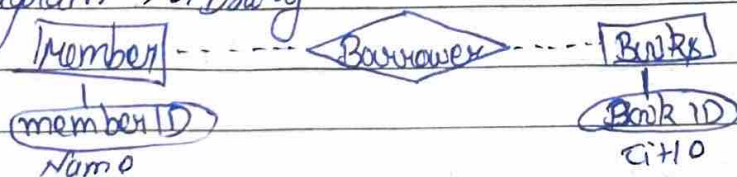
Q7 What are integrity constraints?

- These are rules that ensure correct & valid data.
- Ex: Primary key, foreign key, Unique, Not null.

Q8-B Design, Draw & Defend.

- 1) Three-schema Architecture diagram
 - External level
 - Conceptual level
 - Internal level

2) ER diagram → Library

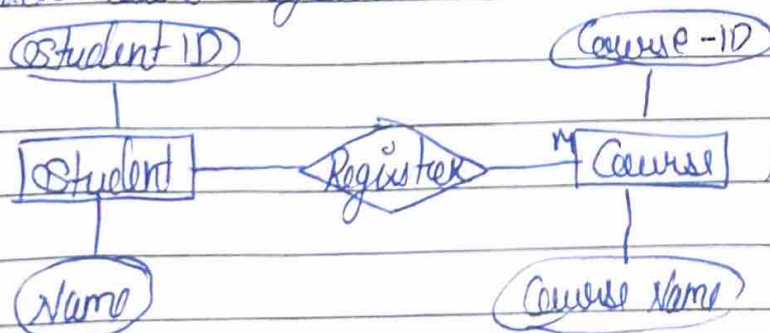


- Entities: Member, Book
- Relationship → Borrower

→ PK - member ID, Book ID

One member can borrow many books

3) Student Course Registration ER



Student (Student ID, Name)
Course (Course ID, Course Name)
Relationship: student registers for course.

Sec - C Case Based analytical Ques :

1) ER Diagram: Hospital system

Entities: Patient, Doctor, Appointment, Patient books appointment, Doctor attends appointment.

Primary keys: Patient-ID, Doctor-ID, Appointment-ID

ii) Online shopping

- Customer (customer ID - PK)
- Product (Product ID - PK)
- Order (Order ID - PK)

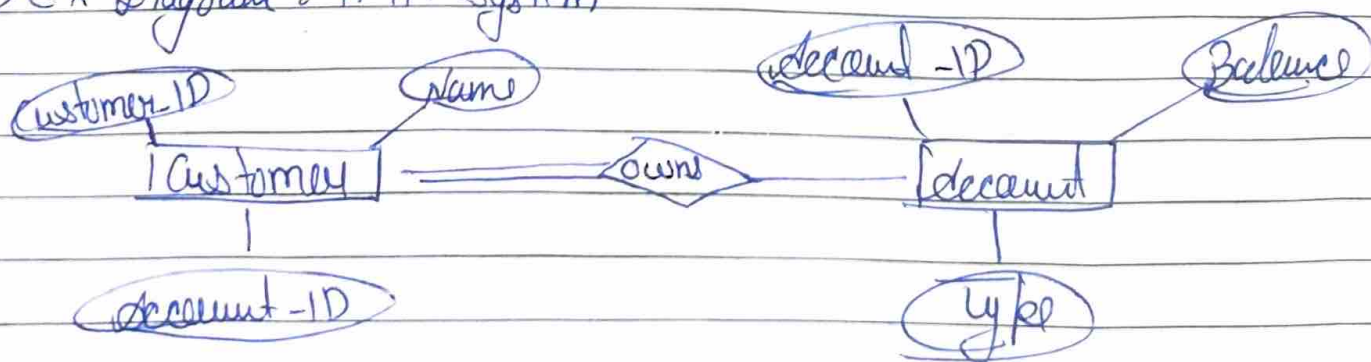
Foreign key in order table connect customer & Product

iii) University - Data abstraction:

- Student see marks (view level)
- Admin manager records (logical level)
- DBA controls storage & (Physical level)

Sec - D One diagram + on Insign

i) ER-Diagram: ATM System



acc-no is Primary key

RM: Customer owns Account

acc-no takes uniquely identifies account.

11) Three - Schema Architecture + One Benefit

Benefit: Provide Data Independence, change in storage does not affect users

Diagram External level

↓
Conceptual level

↓
Internal level

111) Draw an Entity with Attribute

Entity: Employees

Emp-ID (PK)

Name

Salary

Department

→ Primary key

Because for all Employees their ID will be different.